

Sample Questions for EECS6412 Final Exams

(Instructor: Aijun An, York University, Copyright Reserved)

1. Indicate whether each of the following statements is TRUE or FALSE.
 - a. Deep learning algorithms learn deep decision trees.
 - b. What is considered a stopwords depends on the context of the textual data (i.e., the application domain)
 - c. Activation functions like sigmoid and tanh can lead to the gradient vanishing problem due to their saturating nature.
 - d. Dropout is a technique used to increase the complexity of a neural network during training.
 - e. Gradient clipping can prevent exploding gradient problems.
 - f. Noise and outliers can significantly affect the performance of density-based clustering algorithms
 - g. Naive Bayes classifiers can only be used for binary classification tasks.
 - h. k-NN can be used for both classification and regression.
 - i. Complete linkage in hierarchical clustering tends to produce compact, spherical clusters.
2. Multiple choice questions. Only one answer is correct for each question. Circle the right choice.
 - a. Which of the following is not a standard component of the confusion matrix for binary classification?
 - 1) True Positive
 - 2) False Negative
 - 3) Sensitivity
 - 4) True Negative
 - 5) False Positive
 - b. Which of the following is NOT true about AdaBoost?
 - 1) It is an iterative algorithm that sequentially learns a set of weak decision trees.
 - 2) The weak decision trees it learns are independent of each other.
 - 3) It is an ensemble learning method.
 - 4) It gives more weights to examples misclassified by previous models.
 - 5) It learns decision stumps as weak models.
 - c. What is a key disadvantage of the k-means algorithm?

- 1) Works only with categorical data
 - 2) Requires hierarchical structure
 - 3) Sensitive to outliers
 - 4) Uses medoids instead of means
- d. In hierarchical clustering, which linkage method is best suited for detecting elongated, chain-like clusters?
- 1) Complete linkage
 - 2) Single linkage
 - 3) Average linkage
 - 4) Centroid linkage
- e. Which "V" in Big Data highlights the importance of extracting useful insights and business impact from data?
- 1) Variety
 - 2) Velocity
 - 3) Veracity
 - 4) Value
 - 5) Volume
- f. The Apriori algorithm is based on which of the following properties?
- 1) Monotonicity
 - 2) Downward closure property
 - 3) Bayesian inference
 - 4) Hashing property
 - 5) Associative property
- g. A company analyzing customer purchase behavior finds the following rules:
- $\{A\} \rightarrow \{B\}$ (confidence = 90%, lift = 1.2)
 - $\{B\} \rightarrow \{C\}$ (confidence = 80%, lift = 1.1)
 - $\{A\} \rightarrow \{C\}$ (confidence = 75%, lift = 0.8)
- Which rule should be removed from further analysis?
- 1) $\{A\} \rightarrow \{B\}$
 - 2) $\{B\} \rightarrow \{C\}$
 - 3) $\{A\} \rightarrow \{C\}$
 - 4) None of the rules should be removed
- h. What is the purpose of dropout in neural network training?
- 1) To speed up the training process
 - 2) To increase the number of parameters in the network
 - 3) To decrease the learning rate during training
 - 4) To prevent the model from overfitting the training data
 - 5) To reduce the noise level in the training data

- i. What type of output activation function is typically used for a multiclass classification problem?
 - 1) Sigmoid
 - 2) ReLU
 - 3) Tanh
 - 4) Softmax
 - 5) Identity
- j. Which of the following is NOT true about hierarchical clustering methods?
 - 1) It can work in a bottom-up way.
 - 2) It can generate a tree of clusters.
 - 3) Average linkage can be used as a similarity measure in its clustering process.
 - 4) It needs the user to specify the number of clusters to be identified.
 - 5) It takes more time than k-means.

3. Short-Answer Questions

- a. What are the advantages and disadvantages of leave-one-out evaluation?
- b. What are the benefits of using a neural network-based text embedding method over the traditional bag-of-the-words method for text representation?
- c. Consider the following one-dimensional data set with 9 training examples:

x	0.5	2.0	4.4	4.6	4.9	5.2	5.3	5.5	7.0
y	-	-	+	+	+	-	-	+	-

where y is the class attribute with two classes, + and -. Which class is the data point $x = 5.0$ classified into if the 5-nearest neighbor classifier is used?
- d. Discuss how statistical significance testing helps in model evaluation.
- e. What is the main idea behind pre-training a deep network with autoencoders?
- f. Given the following confusion matrix:

	Predicted: Yes	Predicted: No
Actual: Yes	80	20
Actual: No	10	90

Compute Accuracy, Precision, Recall, and F1-score

4. Given the following set of frequent 3-itemsets:

$\{a, b, c\}, \{a, b, d\}, \{a, b, e\}, \{a, c, d\}, \{a, c, e\},$

$\{b, c, d\}, \{b, c, e\}, \{c, d, e\}$

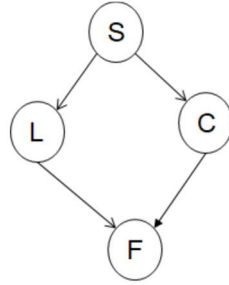
Assume that all the length-3 frequent itemsets in a dataset are listed above. What are the length-4 candidate itemsets generated by the Apriori algorithm? Show the major steps.

5. You are given the following training data to build a Naive Bayes classifier. The task is to classify whether a person will buy a computer (Buy = Yes or Buy = No) based on their age and income.

Example	Age	Income	Buy
1	Young	High	No
2	Young	High	No
3	Middle	High	Yes
4	Senior	Medium	Yes
5	Senior	Low	Yes
6	Senior	Low	No
7	Middle	Low	Yes
8	Young	Medium	No
9	Young	Low	Yes
10	Senior	Medium	Yes

What is the prediction that the naïve Bayes classifier would make for Buy given Age=Young and Income=Medium? Show the major steps of your computation (including what you learned from the data and how you use it to derive a class label for the example).

6. In the following belief network, S, L, C and F are binary random variables:



How many probabilities are needed to complete the belief network? What are they?

7. Consider the Boolean function defined as follows for two binary inputs $x_1, x_2 \in \{0,1\}$:
- $$f(x_1, x_2) = x_1 \wedge x_2$$

Design a single-layer neural network (no hidden layer) that computes this function. Your answer should include:

- 1) The number of neurons.
- 2) The activation function used.
- 3) The weights and bias of the neuron.
- 4) A brief explanation of how the network produces the correct output.